Tutorial 5
Axial Loaded Members

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1. The axially loaded bar $ABCD$ shown in the Fig. 1 is held between rigid supports. The bar has cross-sectional area A_1 from A to C and $2A_1$ from C to D.

   (a) Derive formulas for the reactions R_A and R_D at the ends of the bar.

   (b) Determine the displacements δ_B and δ_C at points B and C, respectively.

   (c) Draw a diagram in which the abscissa is the distance from the left-hand support to any point in the bar and the ordinate is the horizontal displacement δ at that point.

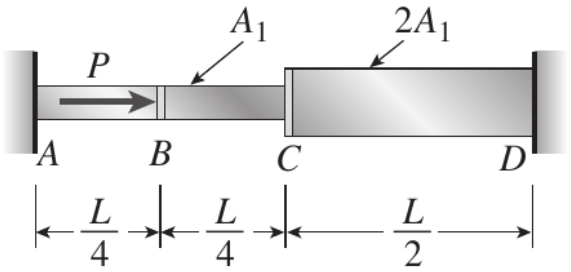


Fig. 1

2. A bar ABC of length L consists of two parts of equal lengths but different diameters. Segment AB has diameter $d_1 = 100\,\text{mm}$, and segment BC has diameter $d_2 = 60\,\text{mm}$. Both segments have length $L/2 = 0.6\,\text{m}$. A longitudinal hole of diameter d is drilled through segment AB for one-half of its length (distance $L/4 = 0.3\,\text{m}$). The bar is made of plastic having modulus of elasticity $E = 4.0\,\text{GPa}$. Compressive loads $P = 110\,\text{kN}$ act at the ends of the bar.

   (a) If the shortening of the bar is limited to 8.0 mm, what is the maximum allowable diameter d_{max} of the hole? (see Fig. 2a)

   (b) Now, if d_{max} is instead set at $d_2/2$, at what distance b from end C should load P be applied to limit the bar shortening to 8.0 mm? (see Fig. 2b)

   (c) Finally, if loads P are applied at the ends and $d_{max} = d_2/2$, what is the permissible length x of the hole if shortening is to be limited to 8.0 mm? (see Fig. 2c)

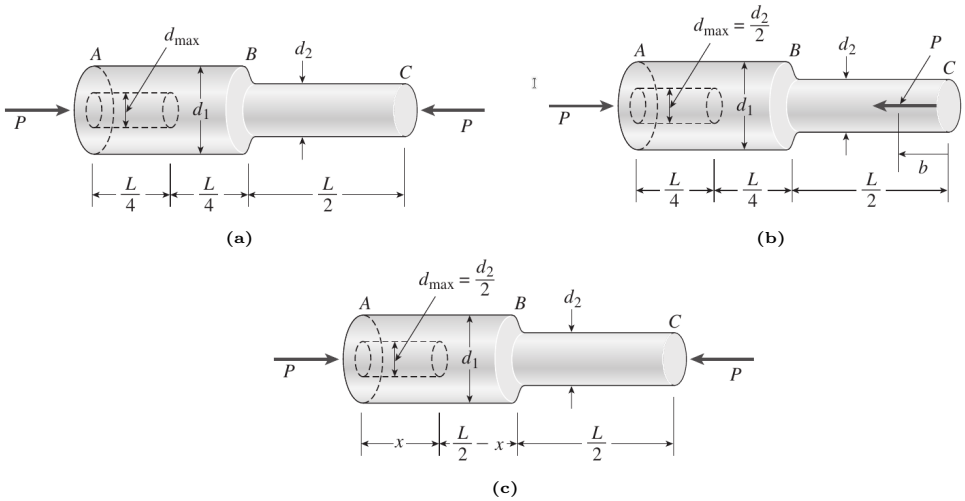


Fig. 2

3. The load of 800 lb is supported by the four 304 stainless steel wires (modulus of elasticity $E = 28(10^3)$ ksi) that are connected to the rigid members AB and DC. Determine the vertical displacement of the load if the members were horizontal before the load was applied. Each wire has a cross-sectional area of $0.05\,\text{in}^2$. (see Fig. 3)

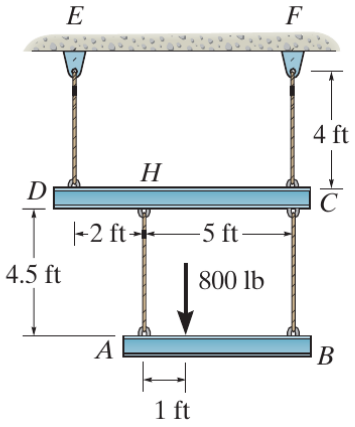


Fig. 3

4. A post AB supporting equipment in a laboratory is tapered uniformly throughout its height H (see Fig. 4). The cross sections of the post are square, with dimensions $b \times b$ at the top and $1.5b \times 1.5b$ at the base. Derive a formula for the shortening δ of the post due to the compressive load P acting at the top. (Assume that the angle of taper is small and disregard the weight of the post itself.)

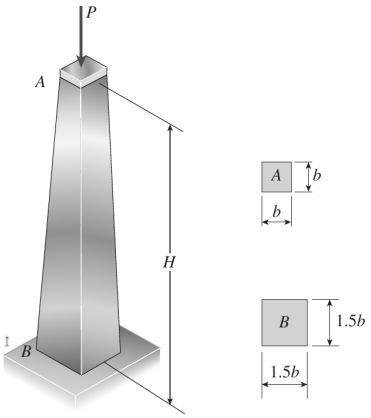


Fig. 4

5. The casting is made of a material that has a specific weight γ and modulus of elasticity E. If it is formed into a pyramid having the dimensions shown in Fig. 5, determine how far its end is displaced due to gravity when it is suspended in the vertical position.

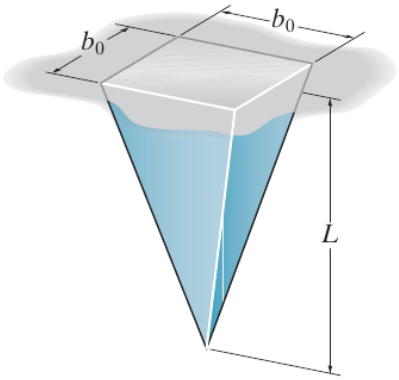


Fig. 5

6. The bolt has a diameter of 20 mm and passes through a tube that has an inner diameter of 50 mm and an outer diameter of 60 mm (see Fig. 6). If the bolt and tube are made of A-36 steel, determine the normal stress in the tube and bolt when a force of 40 kN is applied to the bolt. Assume the end caps are rigid.

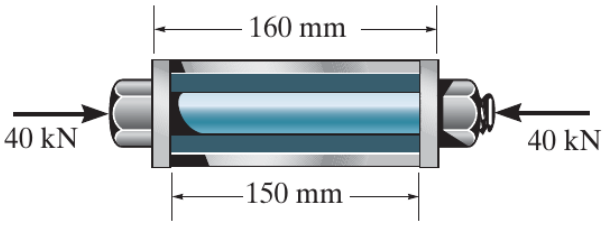


Fig. 6

7. A massive bar as shown in Fig. 7 has Weight(W_o),Cross section area(A), Thermal expansion co-efficient(α_T) is fixed at $x = 0$ and just touches ground in stress free manner.

   (a) Determine the stress distribution $\sigma(x)$ in the bar after uniform heating of ΔT.

   (b) Which ΔT causes compression everywhere in the bar.

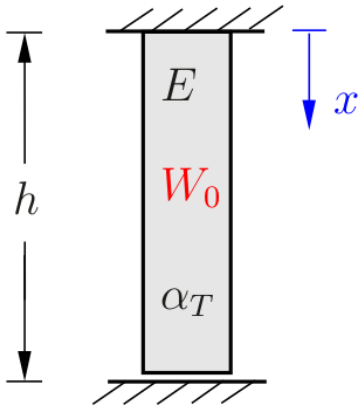


Fig. 7

8. An elastically supported bar as shown in Fig. 8 with $c_1 = 2c_2 = EA/2a$ is loaded by constant axial load n. Compute the distribution of the normal force $N(x)$ in the bar.

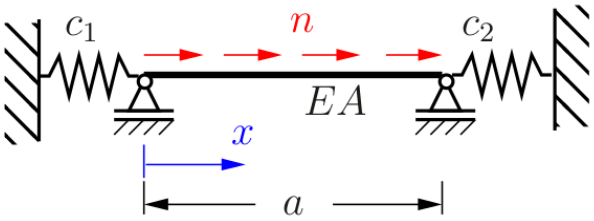


Fig. 8